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Ques. 1 - Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

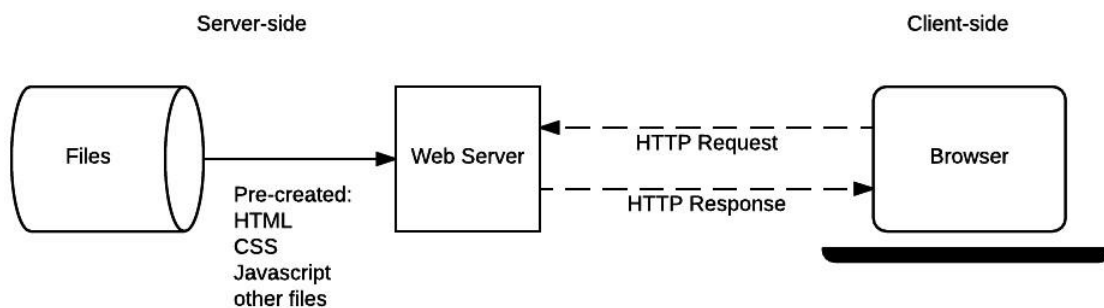
Ans – FrontEnd : Frontend is basically the part of a website or application that users can see and interact. It is like buttons, menus, sidebars, text etc. on webpage. It is built using HTML, CSS, JavaScript and others languages.

BackEnd : BackEnd is the part of website or application that runs behind it like databases, queries, servers, application working logics that handle the flow of data like storage, processing and security.

Full Stack development : Full Stack Development is the process of building an entire software application handling both the user visible interface(Frontend) and the server side logic and databases(backend).

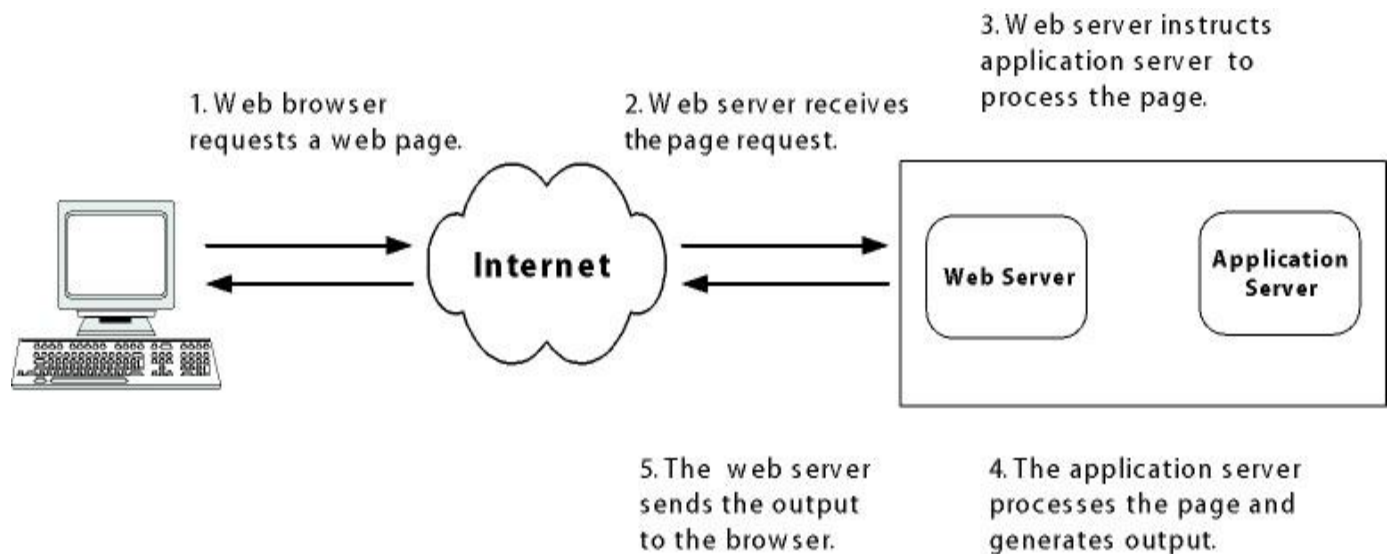
Ques.2 - Create a simple diagram showing how the client-server model works in web architecture.

Ans – This is the simple client server model working diagram.



Ques.3 - Describe how a browser requests and displays a web page from a web server.

Ans – The web browser is an application software used to explore the World Wide Web. It acts as a platform that allows to access information from the internet by serving as an interface between the user and the server. The browser send requests to servers for web documents and services then renders the received HTML files, including text,images, links, styles and scripts.



Ques. 4 - Identify and list the tools required to set up a web development environment. Explain the purpose of each. Ans

– Normal basic tools to start the Web development.

Tool	Purpose
VS Code	Code editor used to write HTML, CSS, JS, etc.
Node.js	Runs JavaScript outside the browser (backend).
Git & GitHub	Version control and code hosting.
Web Browser (Chrome/Edge)	To test and debug your web apps.
Live Server Extension	To preview websites in real-time.

Ques. 5 - Explain what a web server is and give examples of commonly used servers.

Ans – A web server is a combination of hardware and software that stores and delivers web content, like web pages, images and videos to users over the internet when requested by their web browsers using http protocol.

Popular Examples of web servers :

- Apache
- Nginx
- Microsoft IIS
- Node.js (Express)

Ques. 6 - Define the roles of a frontend developer, backend developer, and database administrator in a project.

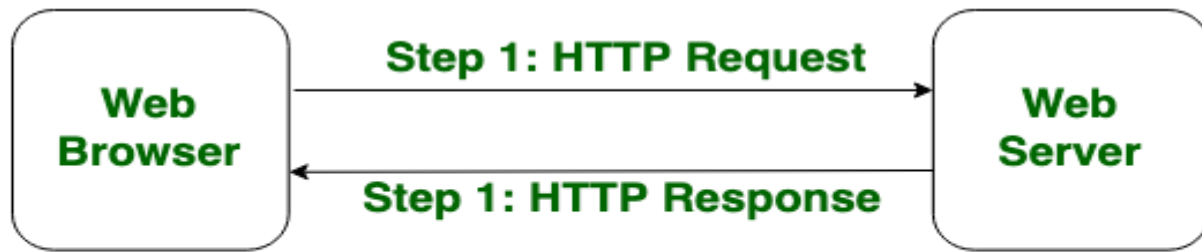
Ans – Front End developer : a frontend developer builds the visual and interactive parts of websites and applications that users see and interact with using HTML, CSS and JavaScript.

BackEnd Developer : a backend developer builds and maintains the invisible components of a website or application, including the server, databases and server side logic.

Database administrator : a database administrator manages and maintains an organization's databases, ensuring they are secure, reliable and accessible.

Ques. 8 - Explain the difference between static and dynamic websites. Provide an example of each.

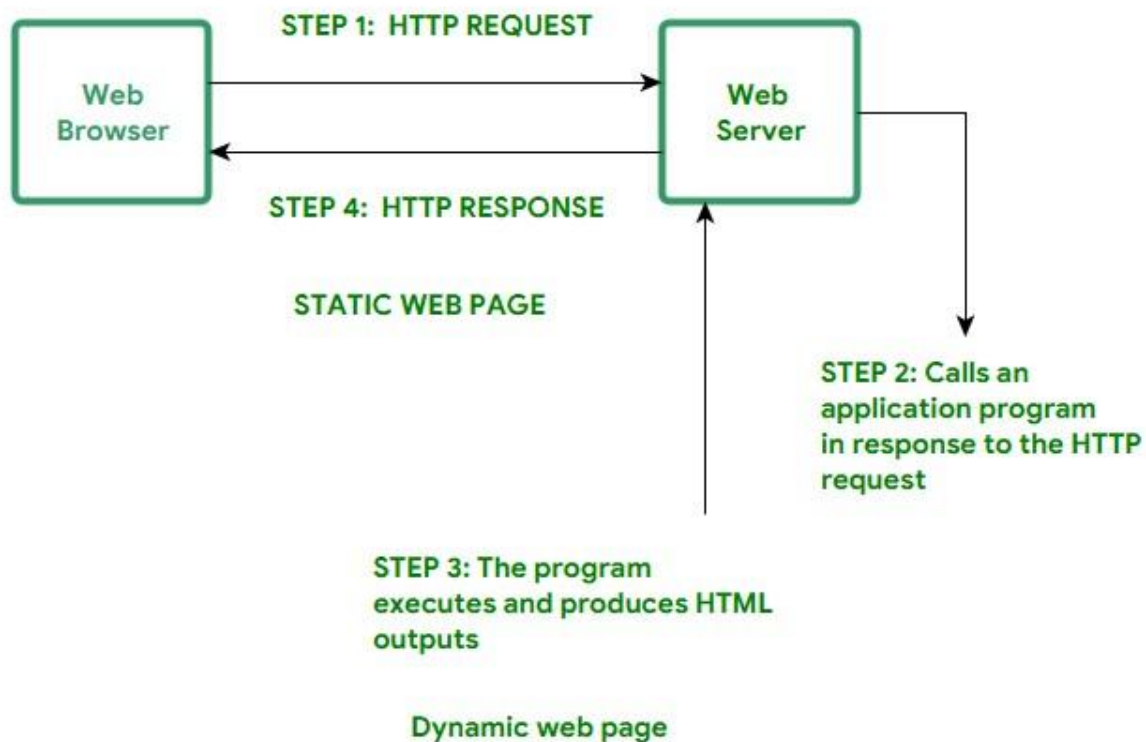
Ans – Static Web Pages : Static web pages are very simple. It is written in languages such as HTML, JavaScript, CSS etc. for static web pages when a server receives a request for a web page, then server sends a response to the client without doing any additional process.



Static Web Page

Dynamic Web Pages : Dynamic web pages are written in languages such as CGI, AJAX, ASP, ASP.NET etc. In dynamic web pages the content of pages is different for visitors. It takes more time to load than the static web page. These web pages are used where the information is changed frequently.

Ex. Stock Prices, Weather Information etc.



Ques. 9 - Research and list five web browsers. Explain how rendering engines differ between them.

Ans -

Browser	Rendering Engine
Google Chrome	Blink
Mozilla Firefox	Gecko
Safari	WebKit
Microsoft Edge	Blink
Opera	Blink

Ques. 10 - Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Ans – Web architecture defines the structure of a website or web application outlining the client, server, network and database components and how they interact each other.

